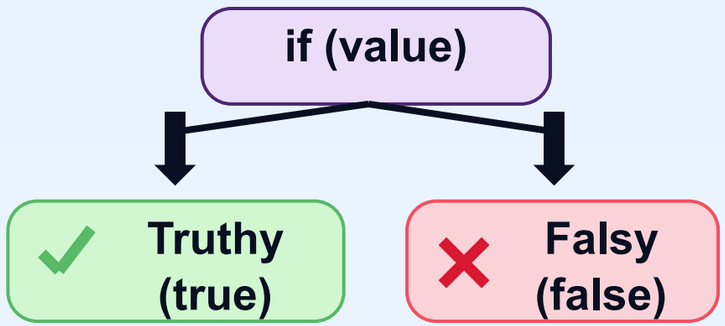
Truthy & Falsy Values

How JavaScript treats values in conditions

What does it mean?

- In a condition, JavaScript converts a value to true or false.
- Truthy values behave like true.
- Falsy values behave like false.



Common Truthy Values

`" "`	Non-empty strings, e.g. "hello"
`#`	Non-zero numbers, e.g. 5, -2, 3.14
`true`	true
`{ }`	Arrays and objects, e.g. [] and {}

```
if ("hello") {
  console.log("truthy");
}
```

Output: truthy

Common Falsy Values

`false`	false
`0`	0
`""`	"" (empty string)
`null`	null
`undefined`	undefined
`NaN`	NaN

```
if (0) {
  console.log("truthy");
} else {
  console.log("falsy");
}
```

Output: falsy

Key Idea

Truthy and falsy values let JavaScript make decisions even when the value is not a real boolean.

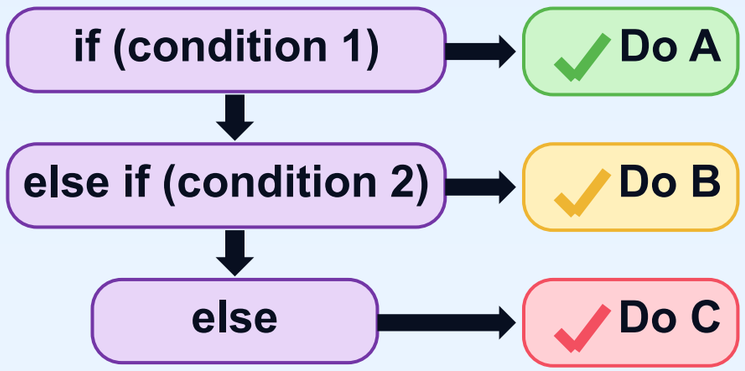
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else if Statement

Checking more than one condition in JavaScript

What is else if?

- Used when the first if condition is false
- Lets you test another condition
- Useful for multiple possible outcomes



How it works

1. Check the first condition
2. If false, check the else if condition
3. Run the matching block
4. Continue to else if no condition matches

```
if (score >= 90) {
  grade = 'A';
} else if (score >= 50) {
  grade = 'Pass';
} else {
  grade = 'Fail';
}
```

Example

score = 75

First condition is false

else if condition is true

```
if (score >= 90) {
  grade = 'A';
} else if (score >= 50) {
  grade = 'Pass';
} else {
  grade = 'Fail';
}
```

Output: Pass

Key Idea

Use else if when your program needs to choose between several conditions.

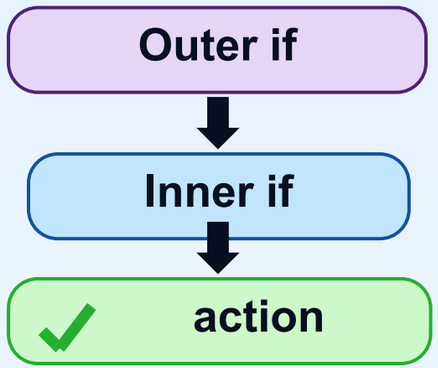
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Nested if Statement

An if statement inside another if statement

What is nested if?

- One if statement is placed inside another
- The inner condition is checked only if the outer one is true
- Useful for detailed decision-making



How it works

1. Check the outer condition first
2. If true, enter the block
3. Then check the inner condition
4. Run the inner action if it is also true

```
if (loggedIn) {
  if (isAdmin) {
    console.log('Open admin panel');
  }
}
```

Example

loggedIn = true

isAdmin = false

User enters outer block but fails inner test

```
if (loggedIn) {
  if (isAdmin) {
    console.log('Admin');
  } else {
    console.log('User only');
  }
}
```

Output: User only

Key Idea

Nested if lets you make a second decision after the first condition succeeds.

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Logical Operators

Combining and reversing conditions in JavaScript

Main Operators

- && means AND.
- || means OR.
- ! means NOT.

&&	||	!
AND	OR	NOT

How they work

&& AND is true when both conditions are true.

|| OR is true when at least one condition is true.

! NOT reverses true to false and false to true.

```
true && false  // false
true || false  // true
!true          // false
```

Example

Check if a student passed and attended.

Use OR for alternative conditions.

```
let passed = true;
let attended = false;
console.log(passed && attended);
```

Output: false

Key Idea

Logical operators connect conditions and make decisions more powerful.